

Student Number

BAULKHAM HILLS HIGH SCHOOL

HIGHER SCHOOL CERTIFICATE

TRIAL EXAMINATION

2008

MATHEMATICS

EXTENSION 1

Time allowed: Two Hours
(Plus 5 mins reading time)

GENERAL INSTRUCTIONS

- Attempt all questions
- There are seven questions - start each question on a new page
- All necessary working should be shown
- Write, using black or blue pen
- Write your student number at the top of each page of the answer sheets
- At the end of the exam, staple your answers in order, behind the cover sheet.

Question 1

- a) A and C have co-ordinates $(-1, 2)$ and $(6, 10)$ respectively.
Find the point B which divides AC internally in the ratio $2 : 3$. 2
- b) Find $\int \frac{4x}{2x+1} dx$ using the substitution $u = 2x + 1$. 3
- c) State the domain of the function $y = \log_e \left(\frac{3x-1}{x+2} \right)$. 3
- d) i) Show that the curves $y = e^{x-1}$ and $y = e^{-x}$ intersect at $x = \frac{1}{2}$. 1
 ii) Find the acute angle between the curves at this point. 3

Question 2 (start a new page)

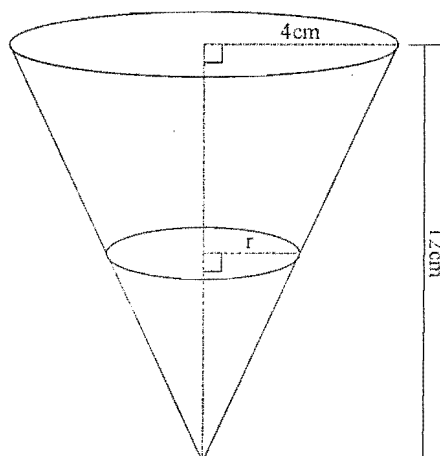
- a) Find the constant term in the expansion $\left(3x^2 + \frac{5}{x^3} \right)^{10}$. 3
- b) Solve $\sin 4x = \cos 2x$ for $0^\circ \leq x \leq 360^\circ$ 3
- c) Evaluate $\int_0^{\frac{3}{4}} \frac{dx}{\sqrt{9-4x^2}}$. 3
- d) Taking $x = 2$ as the first approximation for the root of $\sin x - \frac{x}{3} = 0$ 3
 find a closer approximation of the root using one application of Newton's method.

Question 3 (start a new page)

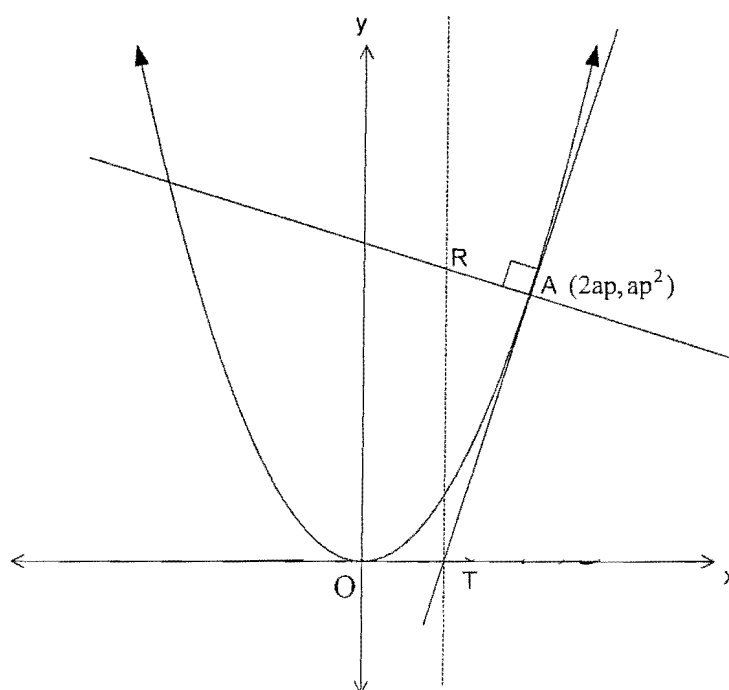
- a) Find the volume when the area between $y = 2 \sin x$, the x and y axes and $x = \frac{\pi}{4}$ is rotated about the x axis. 4
- b) i) State the domain and range of $y = 2 \cos^{-1}(x-1)$ 2
 ii) Hence sketch the curve 2
- c) If α, β and γ are the roots of the cubic $2x^3 - 5x^2 - 3x + 1 = 0$, find
- i) $\alpha + \beta + \gamma$ 1
- ii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ 1
- iii) $\alpha^2 + \beta^2 + \gamma^2$ 2

Question 4 (start a new page)

- a) The diagram shows a conical drinking cup of height 12cm and radius 4cm. The cup is filled with water at a rate of 3cm^3 per second. The height of water at time t seconds is h cm and the radius of the water's surface is r cm.



- i) Show that $r = \frac{1}{3}h$. 1
- ii) Find the rate at which the height is increasing when the height of the water is 9cm. ($V = \frac{1}{3}\pi r^2 h$ is the volume of a cone.) 3
- b) $x = 2at$ and $y = at^2$ are parametric equations for the parabola below.

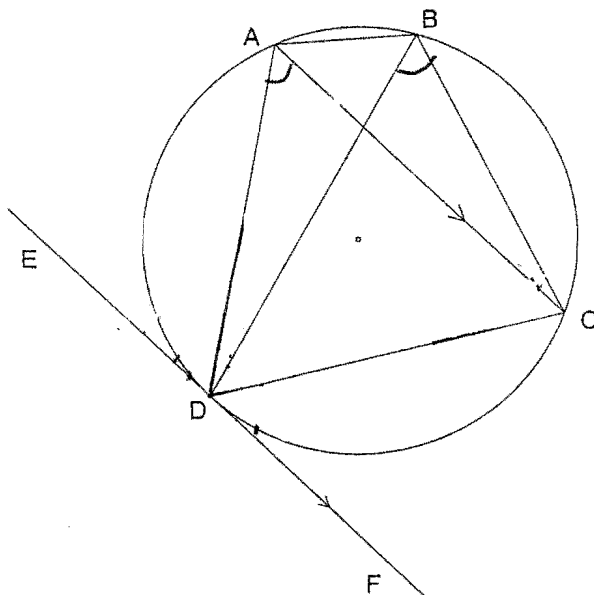


Question 4 (cont.)

- | | | |
|------|---|---|
| i) | By finding the Cartesian equation of the parabola, find the equation of the tangent at the point A. | 2 |
| ii) | The tangent cuts the x axis at T. Find the coordinates of T. | 1 |
| iii) | Find the equation of the normal at A. | 1 |
| iv) | A line through T parallel to the axis of the parabola cuts the normal at R. Show that the coordinates of R are $(ap, ap^2 + a)$. | 1 |
| (v) | Show that the locus of R is a parabola and state the equation of its directrix. | 3 |

Question 5 (start a new page)

- a) ABCD is a cyclic quadrilateral. EF is a tangent to the circle and $AC \parallel EF$. 3



Prove BD bisects $\angle ABC$.

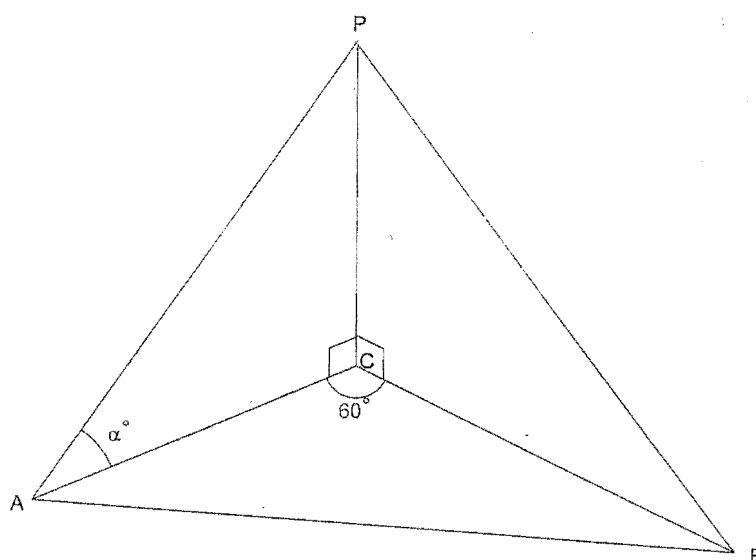
- b) The velocity of a particle as it moves along the X axis is given by

$$v^2 = -9x^2 + 18x + 27$$

- | | | |
|------|--|---|
| i) | Show that the particle undergoes Simple Harmonic Motion. | 2 |
| ii) | What is the period of the motion? | 1 |
| iii) | What is the amplitude of the motion? | 2 |

Question 5 (cont.)

- c) The position of two yachts, A and B out at sea subtend an angle of 60° at the base C of a cliff. The distance AC is 3 times the height of the cliff and the distance BC is 4 times the height of the cliff.



- i) Show that the angle of elevation α° of the cliff from Point A is $18^\circ 26'$ 1
- Ⓐ The distance AB is 300 metres greater than the height of the cliff. Find the height of the cliff. 3

Question 6 (start a new page)

- a) Solve $|x^2 - 9| < 8$ 3
- b) By integrating both sides of the expansion

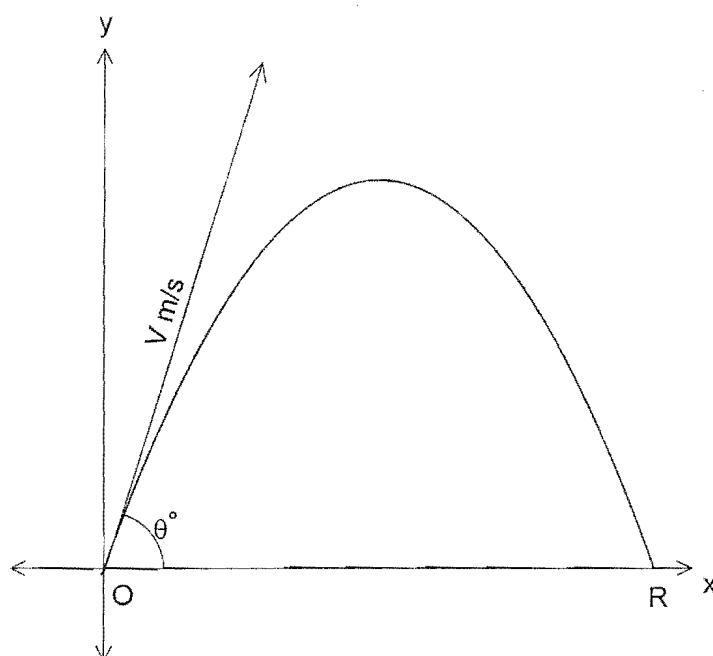
$$(1+x)^n = {}^nC_0 + {}^nC_1x + \dots + {}^nC_nx^n \quad \text{prove} \quad 3$$

$$1 - \frac{1}{2} {}^nC_1 + \frac{1}{3} {}^nC_2 - \dots + \frac{(-1)^n}{n+1} {}^nC_n = \frac{1}{n+1}$$

Question 6 (cont.)

- c) A projectile is fired with velocity V m/s from a point O at an angle θ with the horizontal and hits the ground at a horizontal distance R from O . Taking $g = 10\text{m/s}^2$ you may assume the equations of motion for the projectile.

$$\begin{aligned} \text{i.e. } \quad \ddot{x} &= 0 & \ddot{y} &= -10 \\ \dot{x} &= V \cos \theta & \dot{y} &= -10t + V \sin \theta \\ x &= Vt \cos \theta & y &= -5t^2 + Vt \sin \theta \end{aligned}$$



- i) Show that the range $R = \frac{V^2 \sin 2\theta}{10}$ and that the maximum range is given by $\frac{V^2}{10}$. 3
- ii) The maximum range of a certain rifle is 2000 metres. 3
How much is the range increased when the rifle is mounted on a car travelling at 30 m/s towards the target, the angle of elevation being unaltered.

Question 7 (start a new page)

a) i) Show that $\frac{d}{dx}(\tan^3 x) = 3 \tan^2 x + 3 \tan^4 x$ 2

ii) Hence find $\int \tan^4 x \, dx$ 3

b) Prove by Mathematical Induction that

$$3 \times 2^2 + 3^2 \times 2^3 + \dots + 3^n \times 2^{n+1} = \frac{12}{5}(6^n - 1) \quad \text{for all positive integers } n. \quad 4$$

c) If the 3rd and 4th terms of the binomial $(1 + ax)^n$ are $264x^2$ and $1760x^3$ and $n > a$, find the values of a and n . 3

End of Exam

1a) $(-1, 2)$ $(6, 10)$

$$\left(\frac{2(6)+3(-1)}{2+3}, \frac{2(10)+3(2)}{2+3} \right)$$

$$= \left(\frac{9}{5}, \frac{26}{5} \right)$$

b) $\int \frac{4x}{2x+1} dx$ using $u=2x+1$

$$\frac{du}{dx} = 2 \quad x = \frac{u-1}{2}$$

$$dx = \frac{du}{2} \quad (1)$$

$$\int \frac{4 \left(\frac{u-1}{2} \right) \cdot \frac{du}{2}}{u}$$

$$= \int \frac{u-1}{u} du \quad (1)$$

$$= \int 1 - \frac{1}{u}$$

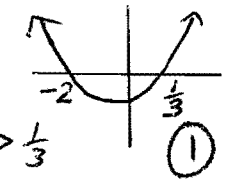
$$= u - \ln u + C \quad (1)$$

$$= (2x+1) - \ln(2x+1) + C$$

(ignore C)

Yr. 12 2008 EXT 1 Trial Solutions

c) $\frac{3x-1}{x+2} > 0 \quad (1) \therefore (3x-1)(x+2) > 0 \quad (1)$



$x < -2, x > \frac{1}{3} \quad (1)$

d) i) $y = e^{-x} \quad y = e^{x-1}$

$$\therefore e^{-x} = e^{x-1}$$

$$\therefore -x = x-1$$

$$2x = 1$$

$$x = \frac{1}{2} \quad (1)$$

ii) $y = e^{-x} \quad y = e^{x-1}$

$$y' = -e^{-x} \quad y' = e^{x-1}$$

at $x = \frac{1}{2} \quad y' = -e^{-\frac{1}{2}} \quad \text{at } x = \frac{1}{2} \quad y' = \frac{1}{\sqrt{e}} \quad (1)$

$$y' = -\frac{1}{\sqrt{e}} \quad y' = \frac{1}{\sqrt{e}} \quad (1)$$

$$\tan \theta = \frac{\frac{1}{\sqrt{e}}}{-\frac{1}{\sqrt{e}}}$$

$$= \frac{2}{\frac{1}{\sqrt{e}}}$$

$$= \frac{2}{1 - \frac{1}{e}}$$

$$\theta = 62^\circ 29' \quad (1)$$

Question 2.

a) $(3x^2 + \frac{5}{x^3})^{10}$

$$T_{k+1} = {}^{10}C_k (3x^2)^{10-k} (5x^{-3})^k \quad (1)$$

$$= {}^{10}C_k 3^{10-k} x^{20-2k} 5^k x^{-3k}$$

$$= {}^{10}C_k 3^{10-k} 5^k x^{20-5k}$$

Constant term $20-5k=0 \quad (1)$

$$k=4$$

Constant term is

$$\therefore T_5 = {}^{10}C_4 3^6 5^4 \quad (1)$$

b) $\sin 4x = \cos 2x \quad 0 \leq x \leq 360^\circ$

$$2 \sin 2x \cos 2x = \cos 2x$$

$$2 \sin 2x \cos 2x - \cos 2x = 0$$

$$\cos 2x (2 \sin 2x - 1) = 0 \quad (1)$$

$$\cos 2x = 0 \quad \sin 2x = \frac{1}{2}$$

$$2x = 90^\circ, 270^\circ, 450^\circ, 630^\circ, 30^\circ, 150^\circ, 390^\circ, 570^\circ$$

$$x = 45^\circ, 135^\circ, 225^\circ, 315^\circ, 15^\circ, 75^\circ, 195^\circ, 285^\circ$$

$(1) \quad (1)$

$$\begin{aligned}
 2c) \int_0^{\frac{3}{4}} \frac{dx}{\sqrt{9-4x^2}} \\
 &= \int_0^{\frac{3}{4}} \frac{dx}{\sqrt{4(\frac{9}{4}-x^2)}} \\
 &= \frac{1}{2} \int_0^{\frac{3}{4}} \frac{dx}{\sqrt{(\frac{3}{2})^2 - x^2}} \quad (1) \\
 &= \frac{1}{2} \left[\sin^{-1}\left(\frac{2x}{3}\right) \right]_0^{\frac{3}{4}} \quad (1) \\
 &= \frac{1}{2} \left[\sin^{-1}\left(\frac{2 \cdot \frac{3}{4}}{3}\right) - \sin^{-1}(0) \right] \\
 &= \frac{1}{2} \left[\frac{\pi}{6} - 0 \right] \\
 &= \frac{\pi}{12} \quad (1)
 \end{aligned}$$

$$d) a_1 = 2 - \frac{f(2)}{f'(2)}$$

$$f(x) = \sin x - \frac{x}{3} \quad f(2) = 0.24 \quad (1)$$

$$f'(x) = \cos x - \frac{1}{3} \quad f'(2) = -0.75 \quad (1)$$

$$\begin{aligned}
 \therefore a_1 &= 2 - \frac{0.24}{-0.75} \\
 a_1 &= 2.3 \quad (1)
 \end{aligned}$$

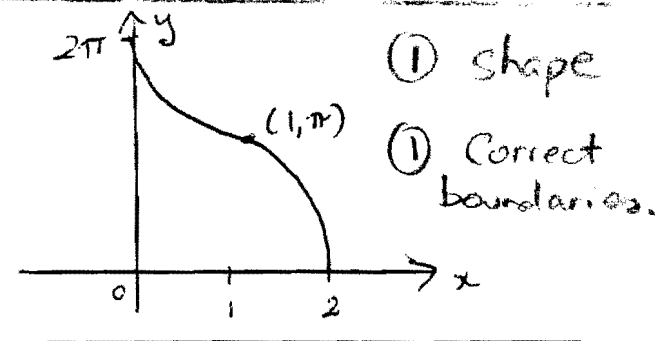
Question 3.

$$\begin{aligned}
 a) V &= \pi \int_0^{\frac{\pi}{4}} 4 \sin^2 x \, dx \quad (1) \\
 &= \pi \int_0^{\frac{\pi}{4}} 4 \left(\frac{1}{2} - \frac{\cos 2x}{2} \right) dx \quad (1) \\
 &= 2\pi \int_0^{\frac{\pi}{4}} (1 - \cos 2x) dx \\
 &= 2\pi \left[x - \frac{\sin 2x}{2} \right]_0^{\frac{\pi}{4}} \quad (1) \\
 &= 2\pi \left[\frac{\pi}{4} - \frac{\sin \frac{\pi}{2}}{2} \right] - [0 - 0] \\
 &= 2\pi \left[\frac{\pi}{4} - \frac{1}{2} \right] \\
 &= \frac{\pi^2}{2} - \pi \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 b) i) y &= 2 \cos^{-1}(x-1) \\
 \frac{y}{2} &= \cos^{-1}(x-1)
 \end{aligned}$$

$$\begin{aligned}
 \text{Domain: } -1 &\leq x-1 \leq 1 \\
 0 &\leq x \leq 2 \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 \text{Range: } 0 &\leq \frac{y}{2} \leq \pi \\
 0 &\leq y \leq 2\pi \quad (1)
 \end{aligned}$$

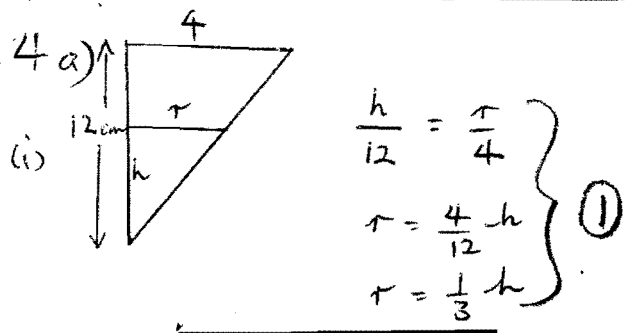


$$c) 2x^3 - 5x^2 - 3x + 1 = 0$$

$$(i) \alpha + \beta + \gamma = \frac{-b}{a} = \frac{5}{2} \quad (1)$$

$$\begin{aligned}
 (ii) \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} &= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} \\
 &= \frac{-\frac{3}{2}}{-\frac{1}{2}} \\
 &= +3 \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 (iii) \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \quad (1) \\
 &= \left(\frac{5}{2}\right)^2 - 2\left(-\frac{3}{2}\right) \\
 &= \frac{37}{4} \quad (1)
 \end{aligned}$$



(i) Find $\frac{dh}{dt}$ when $h=9$.

$$\begin{aligned} V &= \frac{1}{3} \pi r^2 h \\ &= \frac{1}{3} \pi \left(\frac{h}{3}\right)^2 \cdot h \\ &= \frac{\pi h^3}{27} \end{aligned} \textcircled{1}$$

given $\frac{dV}{dt} = 3$

$$\begin{aligned} \frac{dV}{dh} &= \frac{3\pi h^2}{27} \\ &= \frac{\pi h^2}{9} \end{aligned}$$

$$\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$$

$$3 = \frac{\pi \cdot 9^2}{9} \cdot \frac{dh}{dt} \textcircled{1}$$

$$\begin{aligned} \frac{dh}{dt} &= \frac{3}{9\pi} \\ &= \frac{1}{3\pi} \text{ cm/s.} \end{aligned} \textcircled{1}$$

b) $x^2 = 4ay$

(i) $y = \frac{x^2}{4a}$

$$y' = \frac{2x}{4a}$$

must show
① working

at $x=2ap$ $y' = p$ (i.e. $m=p$)

$$\left. \begin{aligned} \text{Tangent } y - ap^2 &= p(x - 2ap) \\ y &= px - ap^2 \end{aligned} \right\} \textcircled{1}$$

(ii) at T $y=0 \therefore 0 = px - ap^2$
 $x = ap$

T(ap, 0) ①

(iii) Eqn of normal

$$y - ap^2 = -\frac{1}{p}(x - 2ap)$$

$$y = -\frac{x}{p} + ap^2 + 2a$$

(iv) at R $x=ap \therefore y = -\frac{ap}{p} + ap^2 + 2a$
 $y = ap^2 + a$ ①

R(ap, ap^2 + a)

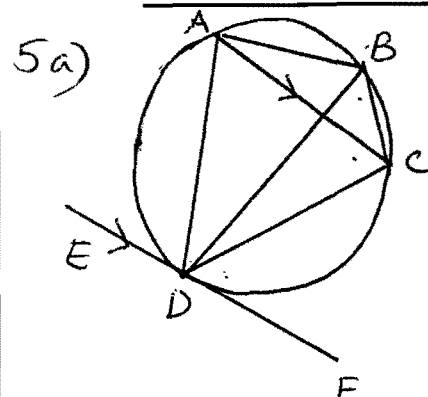
(v) $x=ap \Rightarrow p = \frac{x}{a}$
 $y = ap^2 + a$
 $= a\left(\frac{x}{a}\right)^2 + a$ ①

$y = \frac{x^2}{a} + a$ not essential to state its in the form $y = ax^2 + bx + c$

$$\left. \begin{aligned} x^2 &= ay - a^2 \\ x^2 &= a(y - a) \end{aligned} \right\} \textcircled{1}$$

Vertex (0, a) focal length $\frac{a}{4}$

directrix $y = \frac{3a}{4}$ ①



Let $\angle CDF = x^\circ$

$\therefore \angle DBC = x^\circ$ (Angle between tangent + chord is equal to the angle in the alternate segment) ①

$\angle ACD = x^\circ$ (Alternate $\angle$'s on || lines. ①

$\angle ABD = x^\circ$ (Angles at circumference on the same arc are equal) ①

$\therefore \angle ABD = \angle DBC \therefore BD$ bisects $\angle ABC$.

$$5b) v^2 = -9x^2 + 18x + 27$$

$$(i) \frac{1}{2}v^2 = \frac{-9x^2 + 18x + 27}{2}$$

$$\frac{d}{dx}\left(\frac{1}{2}v^2\right) = -9x + 9 \quad (1)$$

$$\text{ie } \ddot{x} = -9(x-1) \quad \left. \begin{array}{l} \text{which is in the form} \end{array} \right\} (1)$$

$$\ddot{x} = -\omega^2(x-b) \text{ in SHM.}$$

$$(ii) T = \frac{2\pi}{\omega} = \frac{2\pi}{3} \quad (1)$$

$$(iii) \text{ At endpoints } v=0$$

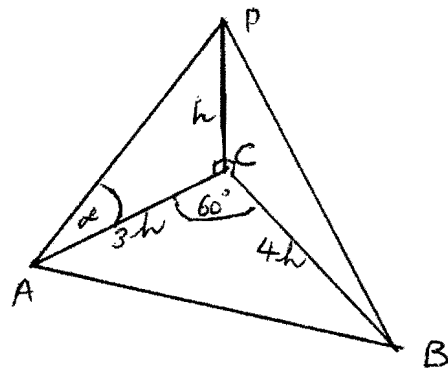
$$\therefore -9x^2 + 18x + 27 = 0$$

$$x^2 - 2x - 3 = 0$$

$$(x-3)(x+1) = 0 \quad (1)$$

$$x = -1, 3$$

$$\therefore \text{Amplitude is } \frac{1 \times 4}{2} = 2 \quad (1)$$



(i)

$$\tan \alpha = \frac{h}{3h}$$

$$\tan \alpha = \frac{1}{3} \quad \alpha = 18.26^\circ \quad (1)$$

$$(ii) (h+300)^2 = (3h)^2 + 4h^2 - 2 \cdot 3h \cdot 4h \cdot \cos 60^\circ \quad (1)$$

$$(h+300)^2 = 9h^2 + 16h^2 - 24h^2 \cdot \frac{1}{2}$$

$$h^2 + 600h + 90000 = 13h^2$$

$$0 = 12h^2 - 600h - 90000 \quad (1)$$

$$= h^2 - 50h - 7500$$

$$h = \frac{50 \pm \sqrt{(-50)^2 - 4 \times 1 \times -7500}}{2 \times 1}$$

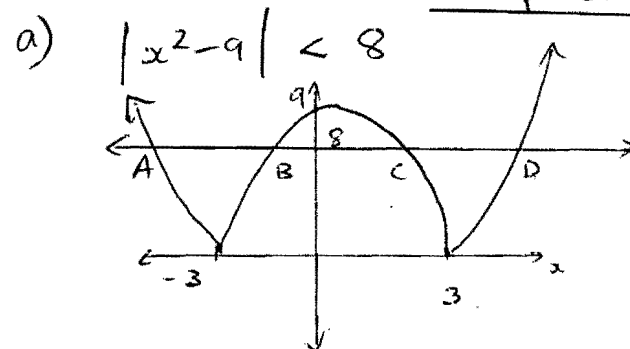
$$= \frac{50 \pm \sqrt{32500}}{2} \quad h > 0$$

$$\therefore h = \frac{50 + \sqrt{32500}}{2} \quad (1)$$

$$= 115.1 \text{ m.}$$

Question 6

Graphically



To find A & D. solve

$$x^2 - 9 = 8$$

$$x = \pm \sqrt{17} \quad (1)$$

To find BC $9 - x^2 = 8$

$$x^2 = 1$$

$$x = \pm 1 \quad (1)$$

$\therefore |x^2 - 9| < 8$ when

$$-\sqrt{17} < x < -1, \quad 1 < x < \sqrt{17} \quad (1)$$

Algebraically.

$$|x^2 - 9| < 8 \quad \Rightarrow \quad -(x^2 - 9) < 8$$

$$x^2 - 9 < 8$$

$$x^2 < 17$$

$$-\sqrt{17} < x < \sqrt{17} \quad (1)$$

$$-(x^2 - 9) < 8$$

$$-x^2 + 9 < 8$$

$$-x^2 < -1$$

$$x^2 > 1 \quad (1)$$

$$\text{ie } x < -1, x > 1$$

require intersection.

$$-\sqrt{17} < x < -1 \quad 1 < x < \sqrt{17} \quad (1)$$

6b)
 $(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_n x^n$

Integrating

$$\frac{(1+x)^n}{n+1} + c_1 = {}^nC_0 x + \frac{1}{2} {}^nC_1 x^2 + \dots + \frac{1}{n+1} {}^nC_n x^{n+1} + c_2 \quad (1)$$

let $x=0$

$$\frac{1}{n+1} + c_1 - c_2 = 0$$

$$c_1 - c_2 = -\frac{1}{n+1} \quad (1)$$

$$\frac{(1+x)^n}{n+1} - \frac{1}{n+1} = {}^nC_0 x + \dots + \frac{1}{n+1} {}^nC_n x^{n+1}$$

let $x=-1$

$$0 - \frac{1}{n+1} = -{}^nC_0 + \frac{1}{2} {}^nC_1 - \frac{1}{3} {}^nC_2 + \dots$$

$$\frac{1}{n+1} = {}^nC_0 - \frac{1}{2} {}^nC_1 + \frac{1}{3} {}^nC_2 + \dots + \frac{(-1)^n}{n+1} {}^nC_n \quad (1)$$

c) Range occurs when $y=0$

$$0 = -5t^2 + Vt \sin \theta$$

$$0 = t(-5t + V \sin \theta)$$

$$t=0, \quad \frac{V \sin \theta}{5} \quad (1)$$

$$x = Vt \cos \theta$$

$$= \frac{V \cdot V \sin \theta \cos \theta}{5}$$

$$= \frac{V^2 \cdot 2 \sin \theta \cos \theta}{10} \quad (1)$$

$$= \frac{V^2 \sin 2\theta}{10}$$

Max range when $\theta = 45^\circ$

$$R = \frac{V^2 \sin 90}{10} \quad (1)$$

$$= \frac{V^2}{10}$$

$$(ii) \quad 2000 = \frac{V^2}{10}$$

$$V^2 = 20000$$

$$V = \sqrt{20000} \quad (V > 0)$$

$$V = 100\sqrt{2} \quad (1)$$

Current horizontal velocity

$$x = V \cos \theta$$

$$= 100\sqrt{2} \cdot \frac{1}{\sqrt{2}}$$

$$= 100$$

$$\therefore \text{New } x = 100 + 30 = 130 \quad (1)$$

$$\text{Currently } y = -10t + V \sin \theta$$

$$= 0 + 100\sqrt{2} \cdot \frac{1}{\sqrt{2}}$$

$$= 100$$

This doesn't change.

$$\therefore V^2 = 100^2 + 130^2$$

$$= 26900 \quad (1)$$

$$\therefore \text{New Range} = \frac{V^2}{10}$$

$$= 2690 \text{ m.}$$

$$\therefore \text{Increase in range} = 690 \text{ m}$$

Question 7.

$$a) (i) \frac{d}{dx} (\tan x)^3 =$$

$$3(\tan x)^2 \cdot \sec^2 x \quad (1)$$

$$= 3 \tan^2 x (1 + \tan^2 x)$$

$$= 3 \tan^2 x + 3 \tan^4 x \quad (1)$$

$$(ii) \frac{d}{dx} (\tan^3 x) = 3 \tan^2 x + 3 \tan^4 x$$

$$\therefore \tan^4 x = \frac{1}{3} \left[\frac{d}{dx} (\tan^3 x) - 3 \tan^2 x \right] \quad (1)$$

$$\int \tan^4 x = \frac{1}{3} \int \frac{d}{dx} (\tan^3 x) - 3 \tan^2 x$$

$$= \frac{1}{3} \left[\frac{d}{dx} (\tan^3 x) - 3(\sec^2 x - 1) \right]$$

$$= \frac{1}{3} [\tan^3 x - 3 \tan x + 3x] + c$$

$$= \frac{\tan^3 x}{3} - \tan x + x + c \quad (1)$$

b) Prove by M.I

$$3 \times 2^2 + 3^2 \times 2^3 + \dots + 3^n \times 2^{n+1} = \frac{12}{5}(6^n - 1)$$

Prove true for $n=1$

$$\text{i.e. } 3^1 \cdot 2^{1+1} = \frac{12}{5}(6^1 - 1)$$

$$3 \times 2^2 = \frac{12}{5} \times 5$$

$$12 = 12$$

$\therefore$ True for $n=1$

Assume true for $n=k$

i.e.

$$3 \times 2^2 + \dots + 3^k \times 2^{k+1} = \frac{12}{5}(6^k - 1) \quad (1)$$

Prove true for $n=k+1$

i.e.

$$3 \times 2^2 + \dots + 3^k \times 2^{k+1} + 3^{k+1} \times 2^{k+2} = \frac{12}{5}(6^{k+1} - 1) \quad (1)$$

i.e.

$$\frac{12}{5}(6^k - 1) + 3^{k+1} \times 2^{k+2} = \frac{12}{5}(6^{k+1} - 1)$$

LHS =

$$\left. \begin{aligned} &\frac{12}{5}(6^k) - \frac{12}{5} + 3 \cdot 3^k \times 2^k \cdot 2^2 \\ &\frac{12}{5}(6^k) - \frac{12}{5} + 12 \cdot 6^k \end{aligned} \right\} (1)$$

$$\frac{2 \cdot 6 \cdot 6^k}{5} - \frac{12}{5} + 2 \cdot 6 \cdot 6^k$$

$$\frac{2}{5} \cdot 6^{k+1} - \frac{12}{5} + 2 \cdot 6^{k+1}$$

$$= \frac{2}{5} \cdot 6^{k+1} - \frac{12}{5} + \frac{10 \cdot 6^{k+1}}{5}$$

$$= \frac{12}{5} \cdot 6^{k+1} - \frac{12}{5}$$

$$= \frac{12}{5}(6^{k+1} - 1)$$

= RHS.

Proved true for $n=1$ & assumed true for $n=k$. Proven true for $n=k+1 \therefore$ true for $n=1, n=2, \dots$ & for all n by M.I.

$$7c) {}^n C_2 a^2 = 264$$

$$\frac{n(n-1)a^2}{2} = 264 \dots (A)$$

$$(1) {}^n C_3 a^3 = 1760$$

$$\therefore \frac{n(n-1)(n-2)a^3}{1 \times 2 \times 3} = 1760 \dots (B)$$

Sub (1) into (2)

$$\frac{264(n-2)a}{3} = 1760$$

$$88(n-2)a = 1760$$

$$(n-2)a = 20$$

$$(n-2) = \frac{20}{a} \quad (1)$$

now n is a pos. integer

$$\therefore a = 1, 2, 4, 5, 10, 20$$

$$p \quad n = 22, 12, 7, 6, 4, 1$$

but $n > a \therefore$

trial $a = 1, 2, 4, 5$.

sub $a=1$ $n=22$ into (A)

$$\frac{22(21) \cdot 1}{2} \neq 264$$

$$a=2 \quad n=12$$

$$\frac{12(11) \cdot 2^2}{2} = 264 \checkmark$$

$$\therefore a=2 \quad n=12 \quad (1)$$

There are other methods of doing this.